SEN 202 – SOFTWARE ENGINEERING
SECURITY INCIDENT MAPPING (SIM) PROJECT ASSIGNMENT

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1. Introduction

Insecurity and community crime are significant problems that can affect students, staff, landlords and other members of a campus community. The assignment proposes a web-based Security Incident Mapping (SIM) platform through which students and villa-mates can quickly report security incidents such as stabbing, sexual violence, theft and murder. The proposed system provides a structured channel for reporting incidents, recording useful information and supporting faster awareness and response. Because security reports may contain sensitive information, the system must emphasize privacy, controlled access and responsible handling of data.

2. Aim and Objectives of SIM

Aim: To design a simple web-based Security Incident Mapping system that improves the reporting, recording, monitoring and management of security incidents within the campus community.

- Provide students, staff and landlords with an accessible platform for reporting security incidents.
- Capture important incident details such as category, description, location and time.
- Provide authorized security personnel with organized reports for review and response.
- Allow users to monitor the status of submitted reports.
- Support location-based awareness of incidents through a map or structured incident list.
- Protect sensitive information by limiting access to authorized users.
- Create a centralized record that can support better documentation and security planning.

3. (a) Questionnaire Design for Google Forms

The questionnaire should be distributed to students, staff and landlords. Its purpose is to understand the nature of security problems, existing reporting methods, user concerns and the features expected from SIM. The form should avoid requesting unnecessary personal information.

No.	Question / Purpose
1	Respondent Category: Which category best describes you? Student / Staff / Landlord.
2	Experience: Have you ever experienced, witnessed or received information about a security incident on or around campus? Yes /
3	Incident Type: Which types of incidents are most common in your area? Theft / Assault / Stabbing / Sexual violence / Vandalism /
4	Location: Where do security incidents occur most often? Campus / Hostel / Residential area / Road or street / Other.
5	Current Reporting: How are incidents currently reported? Security personnel / Police / School authority / Landlord / Informal comm
6	Reporting Difficulty: How easy or difficult is the current method of reporting a security incident? Very easy / Easy / Difficult / Very di

No.	Question / Purpose
7	Delay: What are the major reasons for delayed reporting? Fear / Lack of information / No convenient reporting channel / Poor resp
8	SIM Usage: Would you use a web platform such as SIM to report a security incident? Yes / No / Maybe.
9	Features: Which features would you consider most useful? Incident map / Anonymous option / Evidence upload / Report tracking /
10	Privacy: How concerned are you about the privacy of information submitted through a security-reporting platform? Very concerned
11	Location Sharing: Would you be willing to provide the location of an incident to help security personnel respond? Yes / No / Depen
12	Suggestions: What additional feature or improvement would you recommend for a campus security reporting platform?

4. (b) Ishikawa (Fishbone) Problem Analysis

Problem statement: High occurrence or risk of campus/community security incidents combined with slow, unstructured or ineffective reporting and response.

Cause Category	Possible Causes	Effect on the Problem
People	Low awareness; fear of reporting; lack of cooperation; uncertainty about when to contact	Incidents may not be reported or be reported late.
Methods	Unclear reporting procedure; dependence on informal communication; no standardized method	Information is scattered, incomplete, inconsistent or delayed.
Technology	No centralized incident platform; limited digital reporting; lack of	Security personnel may not receive timely or organized information.
Environment	Poorly monitored areas; isolated locations; inadequate lighting or visibility	Certain locations may become more vulnerable to incidents.
Management	Slow coordination; weak follow-up; limited centralized records.	Response and decision-making may be less effective.
Information	Missing time/location details; inaccurate reports; poor documentation	Difficult to verify incidents and identify patterns.

Interpretation: The Ishikawa analysis shows that the problem is not caused by a single factor. People, reporting methods, technology, environment, management and information all contribute. SIM therefore addresses the technology and reporting-method aspects while providing information that can support better coordination.

5. (c) Mini Web Interface – Proposed SIM Solution

The mini interface should remain simple because the assignment advises avoiding unnecessary complexity due to the submission time. The prototype can contain the following main pages/components:

Component	Description
Home/Dashboard	Displays the purpose of SIM, a quick-report button, recent verified incidents and emergency/security contact inform
Report Incident	A form for entering incident category, description, location, date/time and optional supporting evidence.
Incident Map/List	Displays available incident records by location and category. Sensitive details should not be exposed publicly.
My Reports	Allows a logged-in user to view submitted reports and their current status.
Admin/Security Dashboard	Allows authorized personnel to review, verify, update and manage reports.
Login/Registration	Provides user authentication and role-based access.

6. (d) Use-Case Diagram Specification

The following content should be represented graphically in draw.io as a UML use-case diagram. Actors should be placed outside the system boundary labelled “SIM System”, while use cases are placed inside.

Actor	Main Use Cases
Student/Villa-mate	Register/Login; Submit Incident Report; Attach Evidence; View Incident Information; Track Own Report.
Staff	Login; Submit Incident Report; View Appropriate Incident Information; Track Own Report.
Landlord	Login; Submit Incident Report; View Relevant Incident Information.
Security Officer	Login; View Reports; Review/Verify Report; Update Incident Status; Respond/Record Action.
System Administrator	Manage Users; Manage Incident Categories; Manage Reports; Manage Access; Maintain System Records.

Important relationships: “Submit Incident Report” can include entering incident details and optionally attaching evidence. “Review/Verify Report” precedes official status updates where appropriate. The diagram should show that security officers and administrators have greater privileges than ordinary users.

7. (e) Software Requirements Specification (SRS)

7.1 Purpose

This SRS defines the expected behavior and major requirements of the Security Incident Mapping system. It serves as a guide for designing, developing and testing the proposed SIM prototype.

7.2 Scope

SIM will provide a digital channel for reporting security incidents and managing their records. It will support different user roles, incident submission, report review, status tracking and controlled access to sensitive information. The prototype is intended to demonstrate the software-engineering solution rather than replace emergency services or professional security authorities.

7.3 User Classes and Roles

- **Student/Villa-mate:** reports incidents and monitors submitted reports.
- **Staff:** reports incidents and accesses information appropriate to the staff role.
- **Landlord:** reports incidents affecting residential areas or occupants.
- **Security Officer:** reviews, verifies and updates security incidents.
- **Administrator:** manages users, categories, access permissions and system records.

7.4 Functional Requirements

- FR1: The system shall allow authorized users to register and log in.

- FR2: The system shall allow users to submit an incident report.
- FR3: The report shall capture incident type, description, location and date/time.
- FR4: The system should allow supporting evidence to be attached where appropriate.
- FR5: The system shall store submitted reports for authorized retrieval.
- FR6: Security personnel shall be able to review and verify reports.
- FR7: Authorized personnel shall be able to update report status, such as Received, Under Review, Resolved or Referred.
- FR8: Users shall be able to view the status of reports they submitted.
- FR9: The system shall allow administrators to manage users and incident categories.
- FR10: The system shall provide an incident map or structured incident list for appropriate users.
- FR11: The system shall restrict sensitive incident information according to user role.
- FR12: The system shall maintain records needed for accountability and follow-up.

7.5 Non-Functional Requirements

Requirement	Description
Security	The application should use authentication and role-based access so that sensitive reports are not freely accessible.
Privacy	Personal information and sensitive incident details should only be collected when necessary and should be protected from
Usability	The interface should be simple, clear and usable by people with basic web skills.
Performance	Common actions such as opening the dashboard and submitting a report should respond within a reasonable time.
Reliability	Submitted reports should be stored consistently and should not disappear during normal operation.
Availability	The platform should be accessible whenever users need to report incidents, subject to hosting and network availability.
Maintainability	The code and database structure should be organized so that future features can be added without rebuilding the whole sy
Scalability	The design should allow the number of users and incident records to grow without major redesign.

8. Major System Components and Interactions

- **Presentation Layer:** contains the web pages/forms through which users interact with SIM.
- **Authentication and Authorization:** identifies users and controls what each role can access.
- **Incident Management Module:** receives, validates, stores and retrieves incident reports.
- **Mapping/Location Module:** associates an incident with a location for appropriate visualization.
- **Notification/Status Module:** supports status changes and communicates relevant updates.
- **Database:** stores user accounts, incident records, categories, locations and status information.
- **Administration Module:** allows authorized administrators to manage system records and permissions.

9. Sample Incident Data Fields

Field	Purpose
Incident ID	Unique identifier for each report.
Reporter/Reporter ID	Identifies the submitting account where required.
Incident Type	Classifies the reported security event.
Description	Provides a clear explanation of what was reported.
Location	Records where the incident occurred.
Date and Time	Records when the incident occurred or was reported.
Evidence	Optional supporting file or information.
Status	Shows the current stage of the report.
Review Notes	Information added by authorized security personnel.

10. Basic System Workflow

1. User opens SIM and logs in or accesses the appropriate reporting interface.
2. User selects “Report Incident”.
3. User enters the incident category, description, location and date/time.
4. User optionally attaches relevant evidence where safe and appropriate.
5. The system validates the required information and stores the report.
6. An authorized security officer reviews the report.
7. The officer verifies, investigates or refers the report as appropriate.
8. The report status is updated and the user can track the status where permitted.
9. The incident record remains available to authorized personnel for documentation and analysis.

11. GitHub Repository Structure

The project should be organized in a GitHub repository so that the application and its documentation are kept together. A suitable structure is:

SIM-Project/

■■■ README.md

■■■ frontend/ – web interface files

■■■ backend/ – server/application logic, if used

■■■ database/ – database scripts or structure

■■■ documentation/

■ ■■■ SRS.pdf

■ ■■■ Use-Case-Diagram.png

■ ■■■ Ishikawa-Analysis.png

■ ■■■ Questionnaire.pdf

■■■ screenshots/ – prototype screenshots

12. Testing Considerations

Test	Expected Result
Submit valid incident	System accepts and stores the report.
Submit without required fields	System requests the missing information.
Unauthorized access	System denies access to restricted information.
Update report status	Authorized officer can change the status and the new status is saved.
View own report	User can see the status of a report they submitted.
Invalid login	System rejects incorrect credentials.
Administrator management	Administrator can manage permitted users/categories.

13. Conclusion

The Security Incident Mapping system provides a software-engineering solution to the problem described in the assignment. Instead of relying only on informal or disconnected reporting methods, SIM provides a structured platform for collecting incident information and making it available to authorized personnel. The questionnaire helps identify the actual needs of students, staff and landlords, while the Ishikawa analysis identifies major contributing causes. The use-case model defines how different actors interact with the system, and the SRS establishes the functional and non-functional requirements that guide development. With a simple prototype and properly organized project documentation, the system can demonstrate the major stages of a practical software-engineering development process.

Note: The graphical Ishikawa and Use-Case diagrams should be created separately in draw.io and included with the final project documentation.